

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech/MS/PhD Examinations
Assessment - III - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 27 - 04 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. Before attempting, please ensure you have the correct question paper.

Q1	Prove that the derivative of the sigmoid function is a function of the sigmoid function.	[2.5]
Q2	Show that a simple perceptron model ^{can't} generate a nonlinear decision boundary by giving some example.	[2.5]
Q3	Find the SVD of the matrix $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$	[2.5]
Q4	Given the vector $x = [1, 1, 1]^T$, which vector $v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ "best lies in the direction of <u>x</u> " (in the sense of the subspace spanned by the vector, as in the OLS).	[2.5]

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ALL THE BEST

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$\vec{v} \cdot \vec{x}$

cos²